

Full C# Programs with Bugs for Code Review Testing

Program 1 - Student Management

```
using System;
using System.Collections.Generic;

class Program
{
    static void Main()
    {
        List<string> students = new List<string>();
        students.Add("John");
        students.Add("Alice");

        Console.WriteLine(students[5]); // Index out of range

        string name = null;
        Console.WriteLine(name.Length); // Null reference

        int total = 100;
        int count = 0;
        Console.WriteLine(total / count); // Divide by zero
    }
}
```

Program 2 - Employee Salary System

```
using System;

class Employee
{
    public string Name;
    public int Salary;
```

```

    public void PrintDetails()
    {
        Console.WriteLine("Employee: " + Name);
        Console.WriteLine("Salary: " + Salary);
    }
}

class Program
{
    static void Main()
    {
        Employee emp = new Employee();
        emp.PrintDetails();

        int bonus = int.MaxValue;
        bonus += 1; // Overflow

        Console.WriteLine(bonus);
    }
}

```

Program 3 - File Logger

```

using System;
using System.IO;

class Program
{
    static void Main()
    {
        StreamWriter writer = new StreamWriter("log.txt");
        writer.WriteLine("Application Started");

        string text = null;
        Console.WriteLine(text.ToUpper());

        // Missing writer.Close();
    }
}

```

Program 4 - Authentication System

```
using System;

class Program
{
    static void Main()
    {
        string username = "admin";
        string password = "123456";

        Console.WriteLine("Username: " + username);
        Console.WriteLine("Password: " + password);

        string input = "Admin";

        if(input == "admin")
        {
            Console.WriteLine("Login Success");
        }
        else
        {
            Console.WriteLine("Login Failed");
        }
    }
}
```

Program 5 - Infinite Loop Example

```
using System;

class Program
{
    static void Main()
    {
        int i = 0;

        while(i < 10)
        {
            Console.WriteLine(i);
        }
    }
}
```

}
}